

29. The Photoelectric Effect

In this experiment you will study the emission of electrons from a metal surface that is illuminated with light of various discrete frequencies. You will measure the dependence of the kinetic energy of the emitted electrons on the frequency of the incident light.

§1. Introduction

Late in the nineteenth century a series of experiments revealed that electrons are emitted from a metal surface when it is illuminated with light of sufficiently high frequency. This phenomenon is known as the photoelectric effect.

In 1905, Einstein explained the photoelectric effect by assuming that light propagates as individual packets of energy called quanta or photons. This was an extension of the quantum theory developed by Max Planck. In order to explain the spectrum of radiation emitted by bodies hot enough to be luminous, Planck assumed that the radiation is emitted discontinuously as bursts of energy called quanta. Planck found that the quanta associated with a particular frequency ν of light all have the same energy, $E = h\nu$, where $h = 6.626 \times 10^{-34} [\text{J} \cdot \text{s}] = 4.136 \times 10^{-15} [\text{eV} \cdot \text{s}]$ (Planck's constant). Although he had to assume that the electromagnetic energy radiated by a hot object emerges intermittently, Planck did not doubt that it propagated continu-

ously through space as electromagnetic waves. Einstein, in his explanation of the photoelectric effect, proposed that light not only is emitted a quantum at a time, but also propagates as individual quanta.

Einstein's explanation of the photoelectric effect is that it is a result of collisions between photons (light quanta) of the incident light beam and electrons in the metal surface. In the collision, the photon energy $h\nu$ is absorbed by the electron. Some of this energy is then used to overcome the binding energy of the electron to the metal, and the remainder appears as kinetic energy of the freed electron. This quantum theory of light is totally contrary to the wave theory, which predicts that light energy is distributed continuously throughout the wave pattern, and which provides the sole means of explaining many optical effects such as diffraction and interference. This wave-particle duality cannot be avoided; both theories are required to account for the observed behaviour of electromagnetic radiation. The 'true' nature of light cannot be described in terms of everyday experience, and both wave and quantum theories must be accepted, contradictions included, as being closest to a complete description of light.

§ 2. Theory

When light or electromagnetic waves such as X-rays of a specific wavelength hit a metal surface, electrons are emitted from the surface. Although the number of electrons emitted per unit of time increases with the intensity of the incident light, the energy of the electrons remains constant. When the wavelength of the electromagnetic waves is varied, the energy of the emitted electrons is also changed. This phenomenon is called the photoelectric effect and is a key theory for explaining the particle nature of light.

For a given metal, a certain minimum amount of energy is required to separate the electrons bound in the metal from its surface. This minimum required energy, which is called the work function, is different for each type of metal. When a metal is illuminated with light with more energy than the work function $e\phi$, electrons are ejected from the metal's surface. The relationship between the kinetic energy of the ejected electron K_E and the wavelength of light λ is known as Einstein's equation, as follows:

$$h\nu = \frac{hc}{\lambda} = K_E + e\phi, \quad (1)$$

where $\nu = c/\lambda$ and c is the velocity of light. Work functions for typical metals are shown in Table 1; they are usually between 1 and 6 [eV].

In photoelectric effect experiments, a vacuum tube called a phototube is usually used as shown in Fig. 1. The phototube consists of two electrodes enclosed in an evacuated tube. One electrode (the cathode) has a large

Table 1 Work function of metals

	Na	Ba	Au	Pt	W
[eV]	2.28	2.51	4.9	5.32	4.52

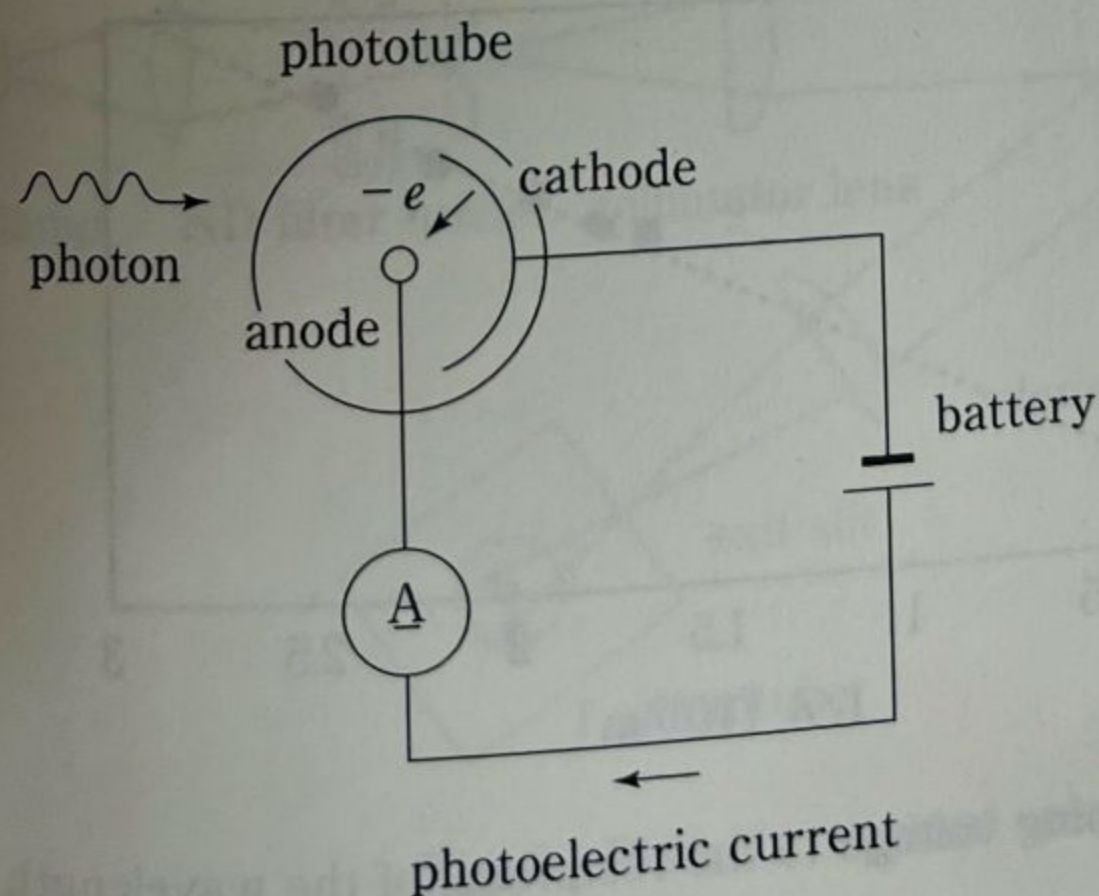


Fig. 1 Phototube

photosensitive surface and is called the emitter. The other electrode (the anode) is a wire and is called the collector. When the emitter is exposed to light, electrons are ejected from its surface. Some of the emitted electrons strike the anode, causing a current to flow. This current is measured by the electrometer. To measure the maximum kinetic energy K_E of these emitted electrons, a retarding potential from a battery is applied across the anode and cathode. The anode is made progressively more negative than the cathode, resulting in fewer and fewer electrons having sufficient kinetic energy to overcome this retarding potential difference. When the anode potential becomes sufficiently large (equalling V_s , the stopping voltage), subsequent photoelectrons have insufficient kinetic energy to overcome the potential difference, so no more electrons reach the anode and the photoelectric current reaches zero. This occurs when

$$K_E = eV_s. \quad (2)$$

Inserting Eq. (2) into Eq. (1), we see that V_s depends on the wavelength of the incident light and the work function.

$$V_s = \frac{hc}{e\lambda} - \phi. \quad (3)$$

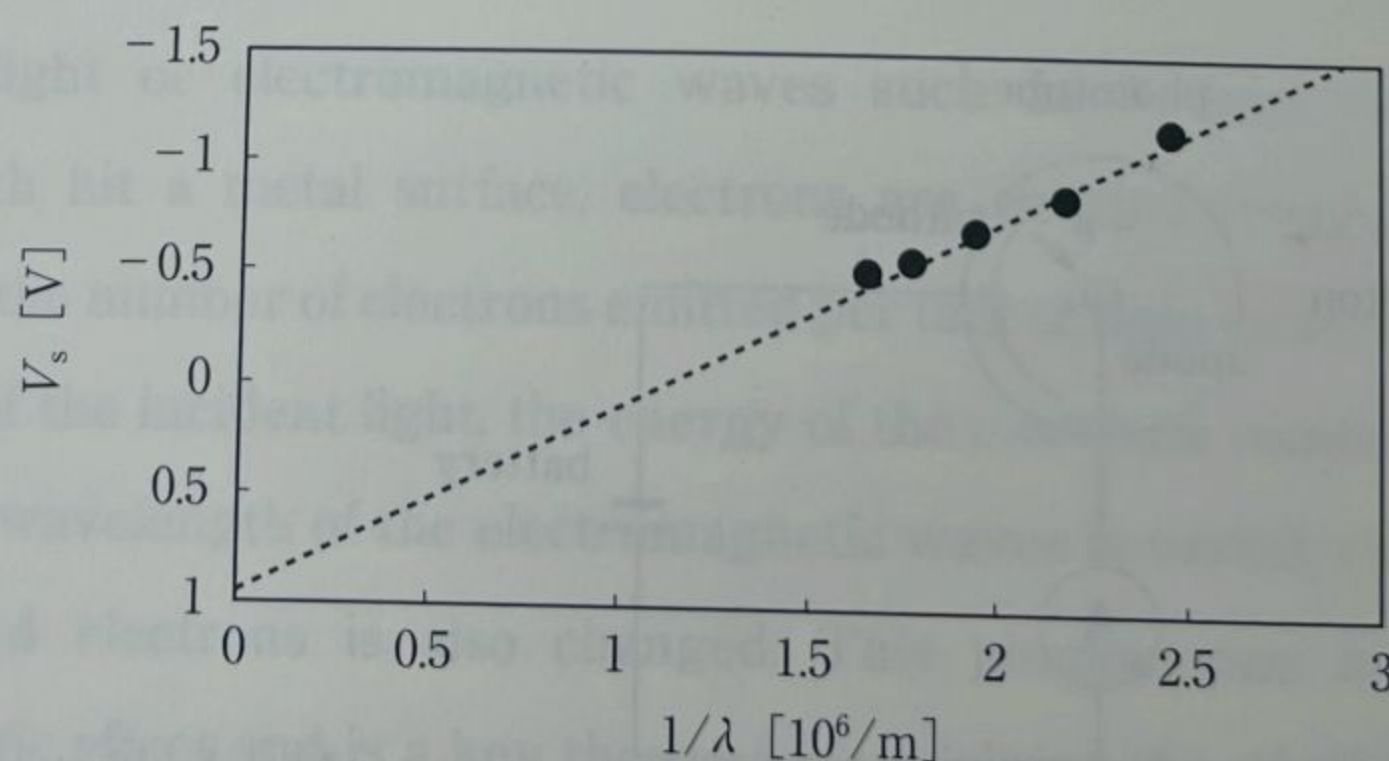


Fig. 2 Dependence of the stopping voltage to the reciprocal of the wavelength of light.

If this stopping voltage is plotted as a function of the reciprocal of the wavelength of light, a straight line is obtained as shown by a dotted line in Fig. 2. The Planck's constant is determined from the slope of the line, while the work function of the cathode metal is obtained from the intersecting point of the straight line with the vertical axis.

§ 3. Experiment

3-1 Equipment

Planck's constant apparatus, Mercury vapour lamp.

The equipment of Planck's constants apparatus shown in Fig. 3 consists of a source of photons (high-intensity halogen lamp source), a monochromator (a diffraction grating), a phototube, an electrometer (essentially a very sensitive ammeter), a DC voltage source for a retarding potential. The diffraction grating enables separation of light from the halogen lamp source into prominent spectral lines of discrete wavelength (and hence frequency), the phototube contains the metal surface to be illuminated, and the

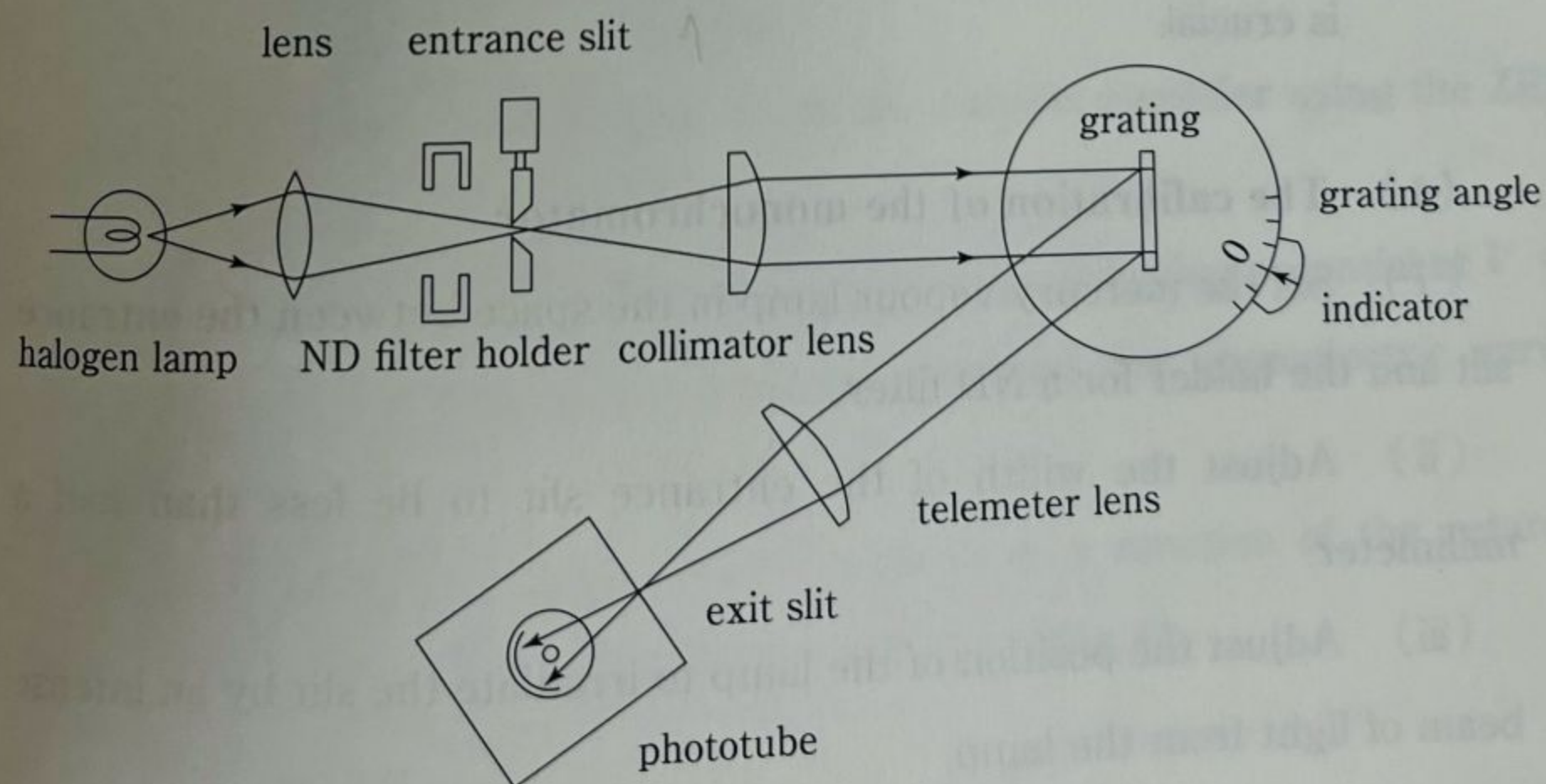


Fig. 3 Planck's constant apparatus

rest of the equipment is used to measure the current due to the emitted photoelectrons as a function of retarding potential.

The vacuum vapour lamp is used to obtain the relationship between the scale showing an angle of the diffraction grating and the wavelength of light emerging from the exit slit. The light from this lamp provides several sharp lines of well known wavelength listed in Table 2.

Table 2 Spectral lines emitted from a mercury vapour lamp [nm]

red	623.4	faint blue	434.75
yellow	579.05	faint blue	433.92
yellow	576.96	violet	407.78
bright green	546.07	violet	404.66
blue	491.6	ultra violet	365
bright blue	435.84		

3-2 Method

NOTE: For best results, this experiment should be done with the room lights off. Also, avoid bumping the equipment as proper alignment is crucial.

(1) The calibration of the monochromator

- (i) Set the mercury vapour lamp in the space between the entrance slit and the holder for a ND filter.
- (ii) Adjust the width of the entrance slit to be less than half a millimeter.
- (iii) Adjust the position of the lamp to irradiate the slit by an intense beam of light from the lamp.
- (iv) Turn the disk fixing the grating and find two bright yellow lines

on a white paper inserted in front of the exit slit.

(v) Remove the paper and observe the photoelectric current flowing through the phototube without any retarding potential from the battery.

(vi) Read the grating angle on the disk when the photoelectric current reaches a peak for each bright yellow line.

(vii) Read the grating angle for each emission line listed in Table 2.

(2) The measurement of the maximum electron energy as a function of the wavelength of the incident light

(i) Remove the mercury vapour lamp from the Planck's constant apparatus.

(ii) Adjust the width of the entrance slit to be half a millimeter.

(iii) Turn on the power of the Planck's constant apparatus and wait for twenty minutes until the emission of the halogen lamp and the operation of the electric circuit are stabilized.

(iv) Move the grating angle of $\theta = 0$ on the disk to the indicator and watch the spectral lines on a white paper inserted in front of the exit slit.

(v) Close the entrance slit and apply the retarding potential of $V = -3$ [V] to the phototube.

(vi) Adjust the balance of the DC current amplifier using the ZERO ADJ. button.

(vii) Remove the white paper and fix the retarding potential at $V = 0$.

(viii) Open slowly the entrance slit until the photoelectric current reaches 100 [μ A].

(ix) Read the photoelectric current as a function of the retarding potential in the range of $V = 0 : -3$ [V] (Fig. 4).

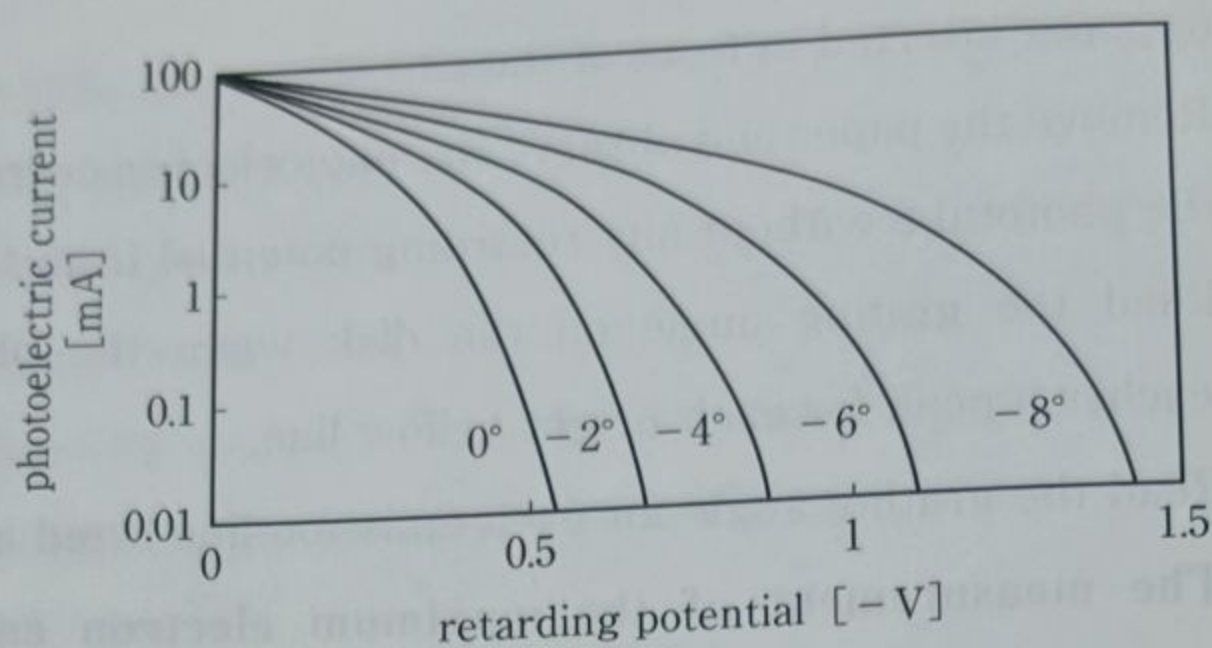


Fig. 4 Dependence of the photoelectric current to the retarding potential.

(x) Repeat the procedures (viii) and (ix) by changing the grating angle in the range of $\theta = -2^\circ : -8^\circ$ at 2° intervals.

§ 4. Analysis

(1) Plot a graph of wavelength of light emitted from the mercury vapour lamp on the y axis versus grating angle on the x axis (Fig. 5).

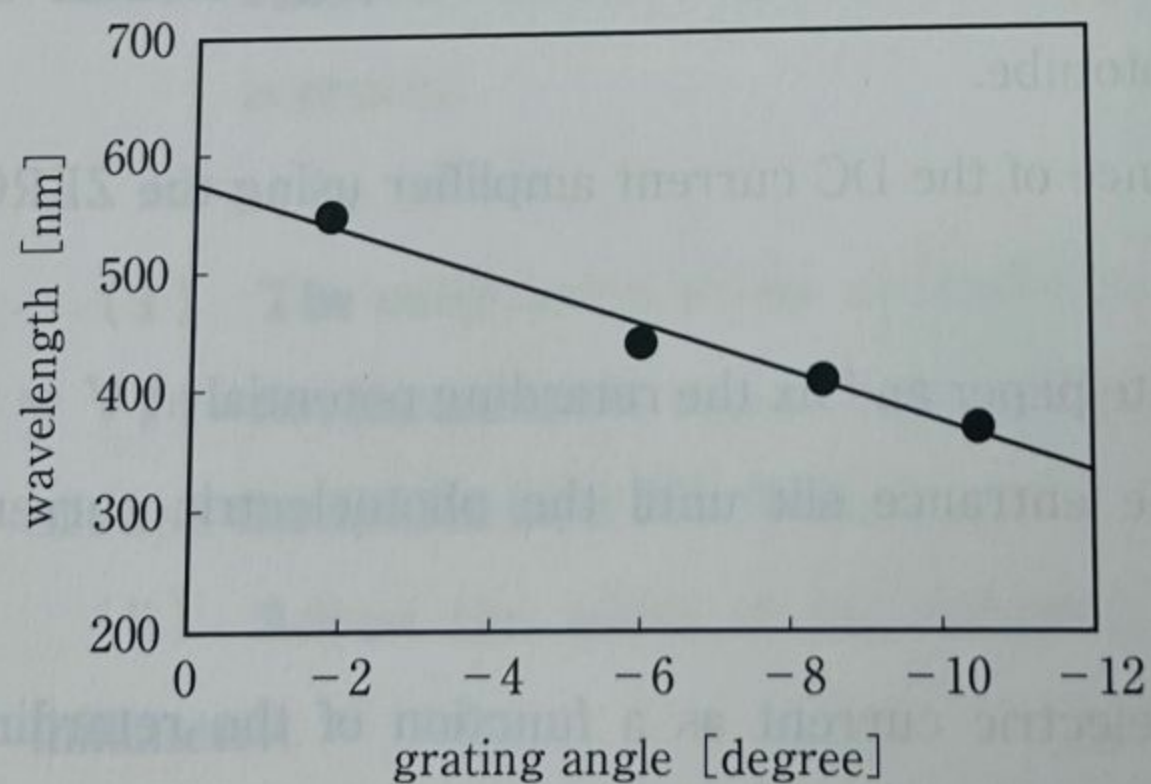


Fig. 5 Calibration of the grating angle

(2) Read the stopping voltage V_s from the graph of the photoelectric current as a function of the retarding potential. Here, the retarding

potential at $0.01 [\mu\text{A}]$ of the photoelectric current is assumed to be the stopping voltage.

(3) Plot a graph of the stopping voltage on the y axis versus wavelength of light on the x axis, which is transformed from the grating angle using the relation shown in Fig. 5 and draw the best-fit line of the data as shown in Fig. 2.

(4) Determine values for the Planck's constant h and the work function $e\phi$ [eV]. And compare your value of h with the accepted value of $6.626 \times 10^{-34} [\text{J}\cdot\text{s}]$ or $4.136 \times 10^{-15} [\text{eV}\cdot\text{s}]$.